

AhamX Bodhi

Where I Becomes Infinite

Multimodal Embedding Space Visualization

Module 4.3: Multimodal Fusion and Training

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Session Overview



What You Will Learn

- What a multimodal embedding space is and why it matters
- How t-SNE and UMAP visualise high-dimensional embeddings
- How to interpret embedding space structure
- What good and bad alignment looks like visually
- Tools for embedding visualisation in practice

Concept at a Glance

- **Duration:** 8 minutes
- **Bloom's Level:** B2 – Understand
- **Difficulty:** L2 – Intermediate
- **Module:** 4.3 Multimodal Fusion and Training
- **Prerequisites:** Multimodal Fusion Strategies, Embeddings

Learning Objectives



By the End of This Session You Will Be Able To

1. **Explain** what a multimodal embedding space is and why alignment is its key property
2. **Interpret** t-SNE and UMAP visualisations of multimodal embeddings
3. **Identify** visual signs of good alignment, modality gap, and cluster structure
4. **Describe** the “modality gap” phenomenon and its implications for retrieval
5. **Apply** visualisation tools (TensorBoard Projector, UMAP-learn) to inspect VLM embeddings
6. **Connect** visualisation insights to model debugging and training quality assessment





What is a Multimodal Embedding Space?

The Core Concept

An **embedding** is a high-dimensional vector that represents some piece of data (an image, a sentence, an audio clip). An **embedding space** is the mathematical space in which all these vectors live. For a model with $d = 512$ dimensions, every image and every sentence is a point in 512-dimensional space. In a **multimodal** embedding space, images, text, and audio all map to vectors in the **same** space, enabling cross-modal comparison via dot product or cosine similarity.

Why the Geometry Matters

- **Semantically similar things should be nearby:** A photo of a cat and the word "cat" should have high cosine similarity
- **Cross-modal retrieval = nearest-neighbour search:** Find the text most similar to a query image
- **Clusters reveal concept structure:** All "cat" images cluster together regardless of cat breed
- **Distances encode relationships:** King - Man + Woman $\approx$ Queen (the classic word2vec demo, now extended to images)



The CLIP Embedding Space: A Shared World Model

CLIP's Shared 512-D Space

After CLIP training, the image encoder and text encoder map their inputs into a **shared 512-dimensional hypersphere** (L2-normalised vectors):

- Image of a golden retriever $\rightarrow$ vector v_{img}
- Text "a golden retriever" $\rightarrow$ vector v_{text}
- Cosine similarity: $v_{\text{img}} \cdot v_{\text{text}} \approx 0.95$ (nearly identical)
- Image of a cat $\rightarrow$ vector far from "golden retriever" text in the space

What This Enables

- **Zero-shot image classification:** Compare image embedding to class name embeddings; take the argmax
- **Image-text retrieval:** Given text query, retrieve nearest image embeddings
- **Visual semantic search:** Search a photo library with natural language
- **Multimodal RAG:** Retrieve relevant images and text for a visual question





Why We Need Dimensionality Reduction

The Challenge: 512 Dimensions

Embedding spaces have hundreds to thousands of dimensions. Humans can visualise at most 3 dimensions. To **inspect, debug, and understand** embedding spaces, we need dimensionality reduction methods that project high-dimensional vectors to 2D or 3D while **preserving the most important structural relationships**.

The Two Dominant Algorithms

- **t-SNE (2008):** Preserves local neighbourhood structure. Each point is surrounded by its closest neighbours in the original space. Good for finding clusters. Not good for preserving global distances.
- **UMAP (2018):** Preserves both local and some global structure. Faster than t-SNE. Better for large datasets. Supports out-of-sample projection (new points can be projected without refit).



Reading a t-SNE Plot of Multimodal Embeddings

What to Look For

When visualising CLIP image embeddings with t-SNE on ImageNet classes:

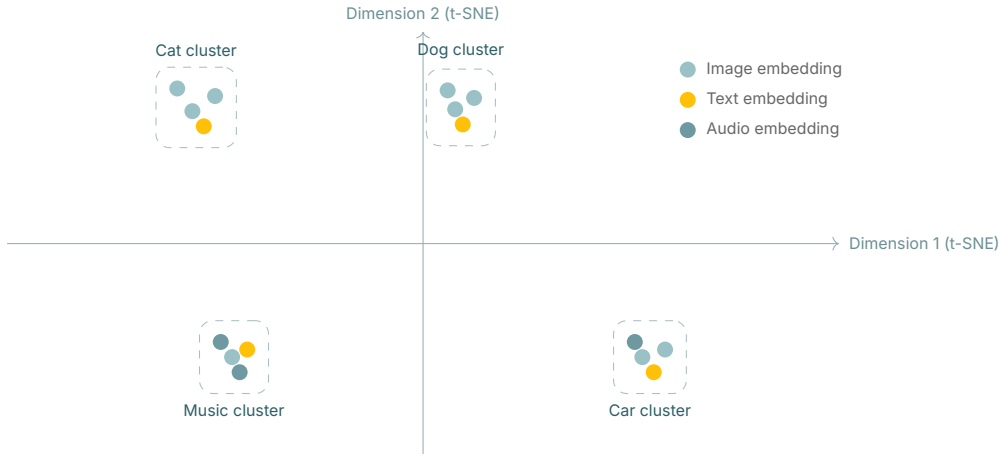
- **Tight clusters** of same-category images = the encoder captures semantic categories
- **Superordinate clusters** (all animals together, all vehicles together) = hierarchical semantic structure
- **Overlapping clusters** = the model cannot distinguish these categories visually
- **Outliers far from clusters** = atypical examples or preprocessing errors

Iconic CLIP Visualisations

- CLIP's t-SNE on ImageNet shows clear class clusters with short-tailed, long-tailed, and aquatic animals forming superordinate clusters
- Text and image embeddings of the same concept appear *close but not identical* in the space
- This small gap between modalities is called the **modality gap** and is a key research topic



Multimodal Embedding Space -- Conceptual Diagram



Key Observation





The Modality Gap: A Surprising Discovery

What Is the Modality Gap? (Liang et al., NeurIPS 2022)

Even in a well-trained CLIP model, image embeddings and text embeddings occupy **different regions** of the hypersphere. If you plot them with t-SNE:

- All image embeddings are in one "hemisphere" of the space
- All text embeddings are in an entirely different "hemisphere"
- Despite both being on the unit hypersphere, there is a consistent gap between modalities
- This gap exists *even after* contrastive training that should align them

Implications of the Modality Gap

- Cross-modal retrieval is comparing across a persistent gap — not all semantics are captured
- The gap is larger for modalities with less paired training data
- Closing the gap requires more aggressive alignment or bridging modules (Q-Former)
- Uniform distribution on hypersphere is not achieved; "cone effect" is observed



Diagnosing Alignment Quality from Embedding Visualisations

What You See in t-SNE/UMAP	Interpretation	What to Do
Tight inter-modal clusters per concept	Good alignment	Model is ready for retrieval
Two separate clouds (images vs text)	Modality gap; poor alignment	More contrastive pre-training or use Q-Former
Mixed cluster with no concept separation	Collapsed representations	Check if contrastive negatives are too easy
Long thin manifolds	Lack of diversity	Diversify training data; increase batch size
Outlier points far from all clusters	Noisy data; OOD samples	Filter noisy pairs; CLIP-score threshold
Domain-specific cluster separated from others	Domain gap	Domain-specific fine-tuning needed





Tools for Embedding Space Visualisation

Production-Ready Tools for AI Engineers

- **TensorBoard Projector:** Built into TensorBoard. Supports t-SNE, UMAP, and PCA. Interactive 3D visualisation. Import embeddings as TSV files with optional labels and images.
- **UMAP-learn (Python):** `pip install umap-learn`. Simple API: `reducer.fit_transform(embeddings)`. Faster than t-SNE; works on millions of points.
- **Weights & Biases (wandb):** Logs embedding tables with images attached. Rich UI for exploring cross-modal neighbourhoods. Integrates with HuggingFace Trainer.
- **Renumics Spotlight:** Dedicated tool for visualising multimodal datasets. Supports images, audio, text, and their embeddings side-by-side.
- **Nomic Atlas:** Web-based embedding exploration at billion-point scale. Free tier available. Supports CLIP embeddings directly.
- **scikit-learn TSNE:** Standard t-SNE for up to 100K points. Use `openTSNE` for larger datasets.



Case Study: Visualising LLaVA's Embedding Space

What Researchers Have Found

When visualising the visual token representations inside LLaVA using UMAP:

- Visual tokens from the same image cluster tightly together in the LLM's token space
- After the MLP projector, visual tokens shift significantly in the embedding space — they become more “text-like”
- The projector acts as a **domain translation layer**: it moves visual representations into the region of the token space that the LLM understands
- Images from similar semantic categories (all animals, all documents, all faces) cluster together even before instruction tuning
- After instruction tuning, concept boundaries sharpen and cross-modal concepts align more tightly

Practical Debugging Use

Visualising embeddings **before and after** fine-tuning reveals whether the projector is learning meaningful cross-modal alignment or just overfitting to task labels.



Nearest-Neighbour Retrieval as Embedding Quality Validation

The Retrieval Test

The most practical way to validate embedding alignment is to perform cross-modal nearest-neighbour retrieval and inspect the results qualitatively:

- Encode 1,000 images and 1,000 text captions
- For each image, retrieve the top-5 nearest text captions by cosine similarity
- For each caption, retrieve the top-5 nearest images
- Manually inspect: are the retrieved items semantically relevant?

Quantitative Metrics

- **Recall@K (R@K):** What fraction of queries have the correct match in the top-K results? (R@1, R@5, R@10 are standard)
- **Median Rank:** What is the median position of the correct match across all queries? Lower is better.
- **COCO Retrieval Benchmark:** 5K images, 25K captions. Standard cross-modal retrieval evaluation.



Practical Embedding Visualisation Workflow

Step-by-Step Guide for AI Engineers

1. **Extract embeddings:** Run your VLM encoder on a balanced sample (500–5,000 items per modality per class) and save the embeddings to `numpy` arrays
2. **Reduce dimensions:** Apply UMAP with `n_components=2`, `n_neighbors=15`, `metric='cosine'` to all embeddings together
3. **Colour by modality and class:** Plot with modality as shape (circle/triangle/square) and semantic class as colour
4. **Inspect:** Look for inter-modal alignment (same class, different modality), intra-modal separation (different classes), and outliers
5. **Run R@K retrieval:** Quantify alignment quality with cross-modal recall metrics
6. **Iterate:** Use insights to guide data curation, projector architecture, and fine-tuning decisions



Summary: Multimodal Embedding Space Visualisation

Key Takeaways

1. A **multimodal embedding space** maps images, text, and audio to the same high-dimensional space, enabling cross-modal comparison
2. **t-SNE and UMAP** reduce embeddings to 2D/3D for visual inspection while preserving neighbourhood structure
3. In a well-aligned space, **semantic clusters span modalities**: an image of a cat and the word "cat" appear near each other
4. The **modality gap** shows that even CLIP keeps image and text embeddings in separate regions — a known limitation under active research
5. **Nearest-neighbour retrieval** and **Recall@K** provide quantitative validation of embedding alignment quality
6. Tools like **TensorBoard Projector**, **Nomic Atlas**, and **Renomics Spotlight** make embedding exploration accessible and interactive

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